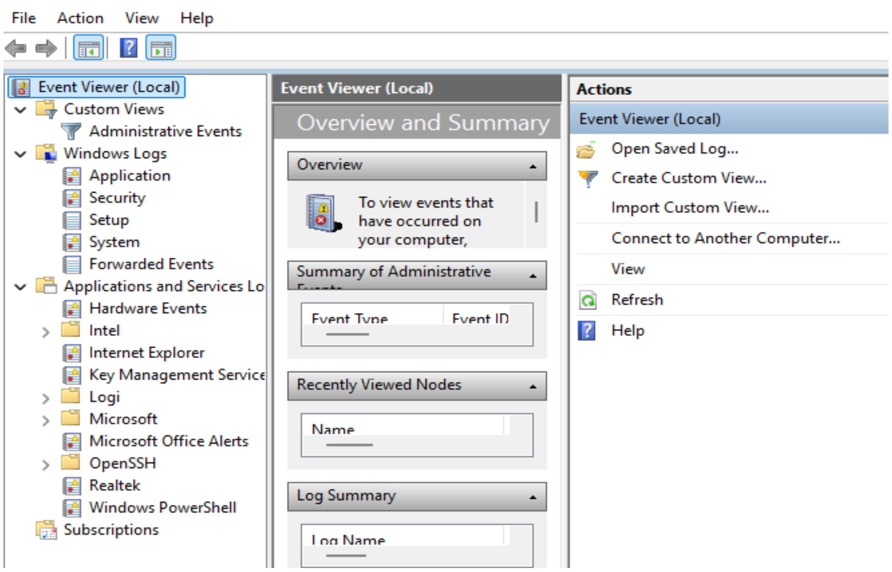
Project 2: Windows Event Log Investigation

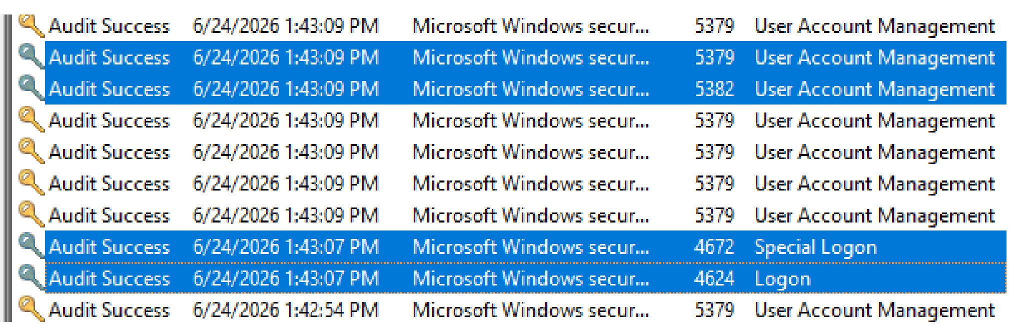
My Findings:

I wanted to take some time to explore my own personal Windows Event Viewer. I knew that a few common tags to look for were 4624, 4625 and 4688. So, I decided to pull it up on my desktop and see what they looked like.

Upon first opening my Event Viewer I noticed there were several more logs than I had originally expected. In the future I would like to explore additional areas like the Key Management Service, Hardware Events, and OpenSSH specifically. Today however, I was going to stick to the security logs.



The first ID I ran into was 5379, in fact a huge majority of the events logged had this ID. Ok, let me click into the event and see if it provides more information.



Audit Success	6/24/2026 1:43:09 PM	Microsoft Windows secur...	5379	User Account Management
Audit Success	6/24/2026 1:43:09 PM	Microsoft Windows secur...	5379	User Account Management
Audit Success	6/24/2026 1:43:09 PM	Microsoft Windows secur...	5382	User Account Management
Audit Success	6/24/2026 1:43:09 PM	Microsoft Windows secur...	5379	User Account Management
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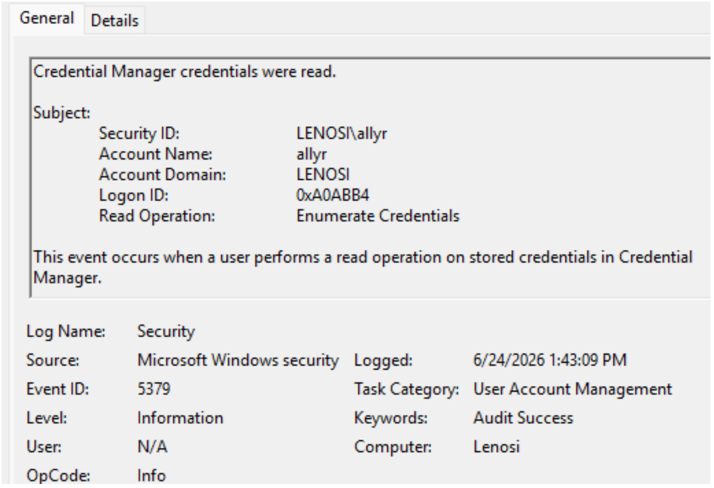
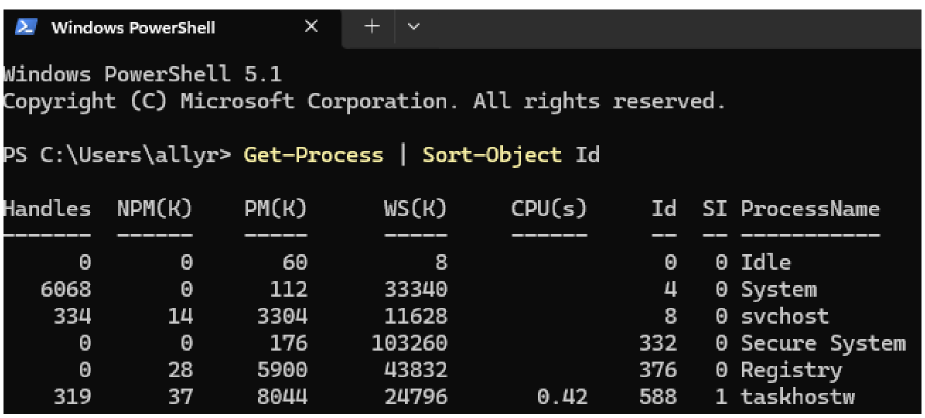
Audit Success	6/24/2026 1:43:09 PM	Microsoft Windows secur...	5379	User Account Management
Audit Success	6/24/2026 1:43:09 PM	Microsoft Windows secur...	5379	User Account Management
Audit Success	6/24/2026 1:43:07 PM	Microsoft Windows secur...	4672	Special Logon
Audit Success	6/24/2026 1:43:07 PM	Microsoft Windows secur...	4624	Logon
Audit Success	6/24/2026 1:42:54 PM	Microsoft Windows secur...	5379	User Account Management

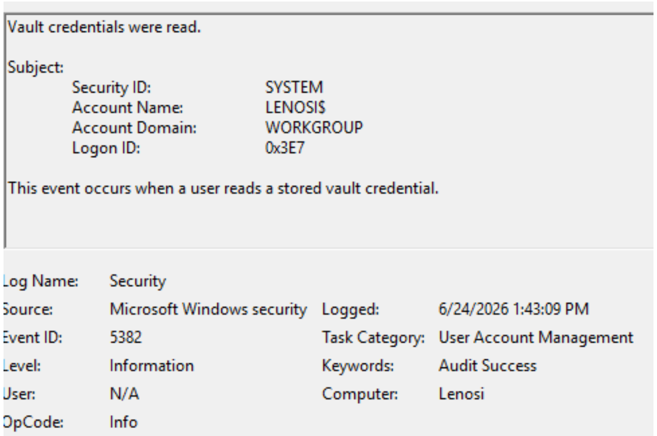
Cool! It has all sorts of information including what that specific event code means. I can see that my account triggered the event and that it accessed a stored credential related to my Microsoft account.



Alright, before I move on from this one, I am curious why there are so many of these events in such a short period. A quick Google search informed me that background services like OneDrive could trigger this event. Unfortunately, I do not see an executable listed for any of these events, but I do see a client process id. I opened PowerShell to see what that client id brought up, unfortunately it was no longer available. I will post a screenshot of a snippet that was currently running at that time, however the list was much longer.



Ok, next on the list was ID 5382, I clicked right in and saw that this event “Occurs when a user reads a stored vault credential”.


Vault credentials were read.

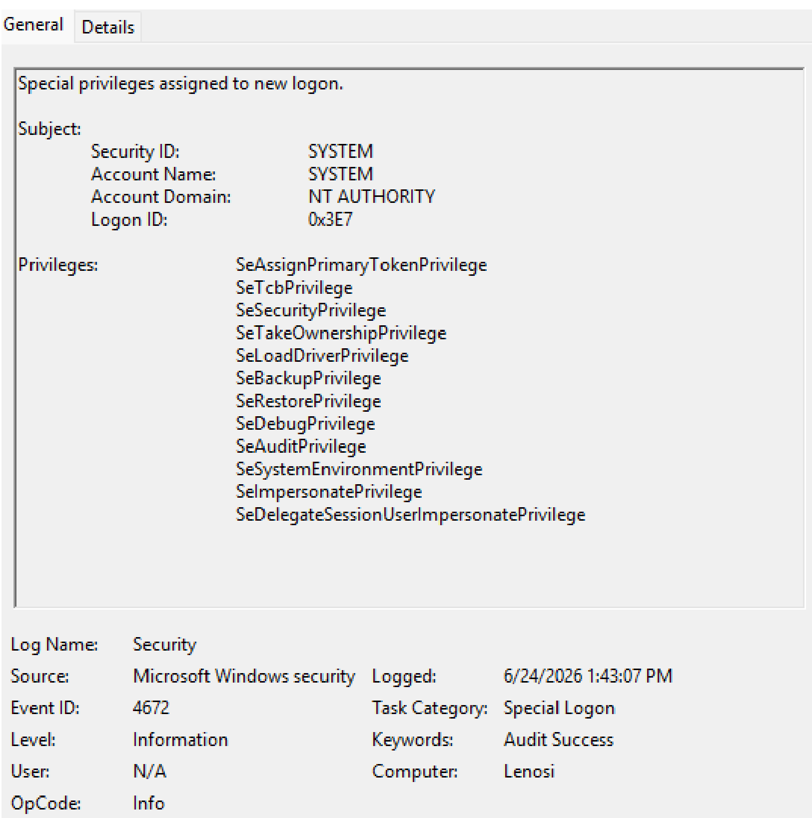
Subject:
Security ID: SYSTEM
Account Name: LENOSI$
Account Domain: WORKGROUP
Logon ID: 0x3E7

This event occurs when a user reads a stored vault credential.

Log Name:	Security		
Source:	Microsoft Windows security	Logged:	6/24/2026 1:43:09 PM
Event ID:	5382	Task Category:	User Account Management
Level:	Information	Keywords:	Audit Success
User:	N/A	Computer:	Lenosi
OpCode:	Info		

I noticed that they both appeared to occur at the exact same time, down to the second. This tells me that they are related events. Because the description for these two events was similar, I wanted to figure out what the actual difference was. What's interesting about this one is that the security ID, account name, account domain are all different. It appears that ID 5379 was for my user account and ID 5382 is the system account, both were likely due to the same authentication event.

On to the next ID, 4672. Alright, I'll be honest, this one looked a bit overwhelming.


General Details

Special privileges assigned to new logon.

Subject:
Security ID: SYSTEM
Account Name: SYSTEM
Account Domain: NT AUTHORITY
Logon ID: 0x3E7

Privileges: SeAssignPrimaryTokenPrivilege
SeTcbPrivilege
SeSecurityPrivilege
SeTakeOwnershipPrivilege
SeLoadDriverPrivilege
SeBackupPrivilege
SeRestorePrivilege
SeDebugPrivilege
SeAuditPrivilege
SeSystemEnvironmentPrivilege
SeImpersonatePrivilege
SeDelegateSessionUserImpersonatePrivilege

Log Name:	Security		
Source:	Microsoft Windows security	Logged:	6/24/2026 1:43:07 PM
Event ID:	4672	Task Category:	Special Logon
Level:	Information	Keywords:	Audit Success
User:	N/A	Computer:	Lenosi
OpCode:	Info		

It appears that a large number of privileges were authorized with this event which automatically triggered suspicion. At the top it says that "Special privileges assigned to new logon", that language makes me nervous. Let me look a little deeper. It appears that this was a request by the windows system account, and that it was only moments before the other two authentication events. I wonder if this chain of events is from when I unlocked my computer. Let me see if the next one can reveal a little more information.

Event ID 4624 was even larger than the one before it. This tag I knew, meant a successful logon, but I did not anticipate it having so much detail.

```
Subject:
        Security ID:                SYSTEM
        Account Name:               LENOSI$
        Account Domain:             WORKGROUP
        Logon ID:                   0x3E7

Logon Information:
        Logon Type:                 5
        Restricted Admin Mode:      -
        Remote Credential Guard:    -
        Virtual Account:            No
        Elevated Token:             Yes

Impersonation Level:                Impersonation

New Logon:
        Security ID:                SYSTEM
        Account Name:               SYSTEM
        Account Domain:             NT AUTHORITY
        Logon ID:                   0x3E7
        Linked Logon ID:            0x0
        Network Account Name:       -
        Network Account Domain:     -
        Logon GUID:                 {00000000-0000-0000-0000-000000000000}

Process Information:
        Process ID:                 0x6c4
        Process Name:               C:\Windows\System32\services.exe

Network Information:
        Workstation Name:           -
        Source Network Address:     -
        Source Port:                -

Detailed Authentication Information:
        Logon Process:              Advapi
        Authentication Package:     Negotiate
        Transited Services:         -
```

Log Name:	Security		
Source:	Microsoft Windows security	Logged:	6/24/2026 1:43:07 PM
Event ID:	4624	Task Category:	Logon
Level:	Information	Keywords:	Audit Success
User:	N/A	Computer:	Lenosi
OpCode:	Info		

However, the really awesome part about this one was at the bottom there was actually a description for much of this information!

This event is generated when a logon session is created. It is generated on the computer that was accessed.

The subject fields indicate the account on the local system which requested the logon. This is most commonly a service such as the Server service, or a local process such as Winlogon.exe or Services.exe.

The logon type field indicates the kind of logon that occurred. The most common types are 2 (interactive) and 3 (network).

The New Logon fields indicate the account for whom the new logon was created, i.e. the account that was logged on.

The network fields indicate where a remote logon request originated. Workstation name is not always available and may be left blank in some cases.

The impersonation level field indicates the extent to which a process in the logon session can impersonate.

The authentication information fields provide detailed information about this specific logon request.
- Logon GUID is a unique identifier that can be used to correlate this event with a KDC event.
- Transited services indicate which intermediate services have participated in this logon request.
- Package name indicates which sub-protocol was used among the NTLM protocols.
- Key length indicates the length of the generated session key. This will be 0 if no session key was requested.

Woah, that is actually so helpful, because I had not seen much of this information before. I was particularly nervous about the “impersonation” field. It was interesting to see what each section meant. I did see that my logon type was 5 and only 2 and 3 were mentioned in the details. When I looked it up, logon type 5 is for service logons, and also that there are 13 different types.. That is research for another day. However, this does answer the question I posed above about this being a user login event (which would be a 2).

I was unable to find any tags for 4688 (process creation), I decided to look up why this was. It turns out that windows requires specific audit policies to be turned on in order for them to be tracked. With that information, I opened CMD as an administrator and used the following line to see what it would return: auditpol /get /subcategory:"Process Creation"

Administrator: Command Prompt

```
Microsoft Windows [Version 10.0.26200.8737]
(c) Microsoft Corporation. All rights reserved.

C:\Windows\System32>auditpol /get /subcategory:"Process Creation"
System audit policy
Category/Subcategory                      Setting
Detailed Tracking
  Process Creation                        No Auditing
```

And it looks like that information was correct! My Event Viewer had not been configured to collect this data.

I think what I found most interesting about this project was that of the event IDs I knew to look for, the top 3 I came across on my own PC were not among them. I am sure that in an enterprise environment this will be a very different looking log with even more IDs to learn.